
Project and Time Plan

FYSM64

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Acknowledgement of AI use

Artificial Intelligence (AI) tools were used in the structuring and writing of this project plan and time plan. The idea and outline of the project was developed by the student and supervisor without the use of AI tools. Figures were created by the student without the use of AI tools. The student and supervisor ensures physical accuracy and validity of the text.

1 Project Plan

Quantum measurement is a fundamental aspect of quantum mechanics and is particularly important for the readout and control of solid-state quantum systems. This project will theoretically investigate quantum measurements of quantum-dot(s) systems coupled to microwave resonators, building on previous work within the research group (Nord, 2026) and in close contact with experimental colleagues in the group of Ville Maisi, where related set-ups have been analyzed. In the experiment, information about the quantum-dot(s) system is obtained by probing a microwave resonator and measuring its reflection coefficient.

The project will initially consider a single quantum dot as a relatively simple system in which the connection between the state of the device and the microwave response can be studied. The main focus will then be shifted towards a double quantum dot (DQD), where the relevant degrees of freedom can form an effective two-level system, or qubit. The aim is to understand how a quantum state can be measured through its coupling to a hybrid microwave system and how the measurement itself affects the quantum system.

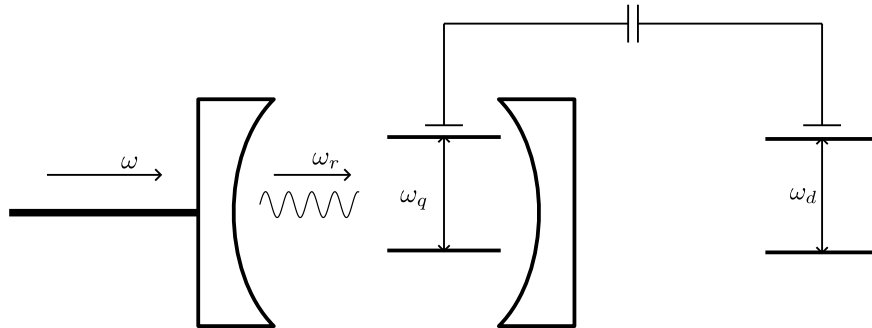


Figure 1.1: The main system that will be considered. The left part shows the hybrid microwave system that will act as a detector. The right part shows a two level system (which will be modelled as a double quantum dot) and the capacitive coupling between the two systems.

A central theoretical framework will be input-output theory, which relates the internal dynamics of a microwave resonator to the experimentally accessible input and output fields. For a resonator coupled to an input field, the output field can, in a simplified description, be written as

$$b_{\text{out}}(t) = b_{\text{in}}(t) + \sqrt{\kappa_c} a(t),$$

where a is the cavity field and κ_c is the coupling rate to the external transmission line. The measured reflection coefficient can consequently be related to the state of the quantum-dot system through its influence on the resonator response.

The coupling between the DQD and the resonator can be described by the Jaynes-Cummings Hamiltonian, or more generally the full quantum Rabi model. Both resonant and dispersive coupling regimes will be considered and are of interest for the experimental group.

An important part of the project will be to understand the measurement process qualitatively, complementing mathematical descriptions that have already been developed within the group. Particular attention will be given to measurement backaction, relaxation and decoherence, and to the conditions under which the measurement can be considered quantum non-demolition (QND). The detector sensitivity and the amount of information obtainable from the measurement will also be investigated. Fisher information will be considered as a possible measure of the information contained in the measured microwave signal about a system parameter ϑ .

A further objective is to investigate the approximations used in the theoretical description and determine under which physical conditions they are valid.

2 Time Plan

- **September:** Literature study on quantum measurement, quantum dots, double quantum dots, microwave resonators, and input-output theory. Familiarization with the experimental setup and the hybrid microwave system. Reproduce basic calculations and results from previous work to establish a starting point for the project.
- **October:** Continue the literature study, with particular focus on input-output theory and the coupling between quantum systems and microwave resonators as well as Fisher information. Develop the theoretical description of the microwave detector and investigate how the reflection coefficient depends on the properties of the coupled quantum-dot system. Extend calculations from previous work and begin outlining the project report.
- **November:** Develop and investigate the model of the coupled quantum-dot and microwave system. Study the single quantum-dot system and then introduce the double quantum dot as an effective two-level system. Investigate the Jaynes-Cummings model and the more general quantum Rabi model, including resonant and dispersive coupling. Perform initial numerical simulations and continue writing the report.
- **December:** Focus on the measurement process and the connection between the quantum state and the experimentally measured reflection coefficient. Investigate measurement backaction, relaxation, and decoherence. Analyse the most relevant results, create figures, and prepare the results and theoretical background for the half-time meeting.
- **January:** Half-time meeting. Assess the progress and results obtained during the first semester and determine the focus of the remaining work. Refine the theoretical model and research questions based on the results obtained so far. Begin a more detailed investigation of QND measurement.
- **February:** Perform the main calculations and simulations concerning quantum measurement of the DQD. Investigate the effects of relaxation and decoherence on the measurement and compare resonant and dispersive coupling regimes. Study detector sensitivity and investigate Fisher information.

- **March:** Analyse the validity of the approximations used in the theoretical description and determine the physical regimes in which they are applicable. Compare different theoretical descriptions where appropriate, and compare the theoretical results with available experimental data. Finalize the main analysis and create figures presenting the central results.
- **April:** Consolidate and interpret the results. Complete the discussion of measurement backaction, decoherence, QND measurement, detector sensitivity, and the validity of the approximations. Write the remaining sections of the report and revise the report as a whole.
- **May:** Finalize the report, including proofreading, figures, references, and conclusions. Prepare slides for the thesis defence and perform a dry run of the presentation. Thesis defence.

References